Variational Quantum Circuits for Pesticide Environmental Fate Prediction: A Comparative Study with Classical Machine Learning on the EU SPIN Database

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Abstract

Predicting pesticide environmental fate properties specifically soil degradation half-life (DegT50) and organic carbon sorption coefficient (K_{oc})—is essential for regulatory risk assessment under EC 1107/2009. Classical QSAR models achieve moderate predictive power for K_{oc} but struggle with DegT50 due to its dependence on complex soil microbial processes. This work investigates whether variational quantum circuits (VQCs) can capture non-linear molecular descriptor relationships that elude classical regressors. We encode 17 physicochemical descriptors for 111 EU-registered pesticides into a 12-qubit IQP-style quantum circuit with data re-uploading and 8 variational layers (301 trainable parameters). Leave-one-out cross-validation benchmarks reveal that Random Forest (RF) achieves $R^2 = 0.194$ for DegT50 and 0.766 for K_{oc}, while Gradient Boosting (GB) reaches 0.223 and 0.779, respectively. The VQC underperforms classical baselines on DegT50 ($R^2 < 0$, 5-fold CV), consistent with severe overfitting expected from the unfavorable parameter-to-sample ratio ($N_{\text{params}}/N_{\text{data}} \approx 2.7$). A hybrid stacking framework blending QML and RF predictions is introduced but remains RF-dominated ($\alpha \approx 0$). We also assess adverse outcome pathway (AOP) features derived from AOP-Wiki data as additional descriptors, finding no improvement over structural features alone. These results provide a rigorous benchmark for near-term quantum machine learning applied to regulatory chemistry and highlight the critical role of dataset size in VQC generalization.

Keywords: quantum machine learning, variational quantum circuits, pesticide fate, DegT50, Koc, QSAR, regulatory chemistry, EU SPIN

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1 Introduction

Pesticide environmental fate assessment is a cornerstone of EU regulatory approval under Regulation (EC) No 1107/2009 (ec1, 2009). Two properties dominate the risk evaluation: the soil degradation half-life (DegT50), governing persistence, and the organic carbon sorption coefficient (K_{oc}), governing mobility and leaching potential. Together, these parameters feed into the FOCUS groundwater models (foc, 2006) that determine whether a substance can be approved for agricultural use.

Experimental determination of DegT50 follows OECD Guideline 307 (oec, 2002), requiring months-long incubation studies under controlled temperature and moisture. K_{oc} is measured via batch equilibrium adsorption (OECD 106) (oec, 2000). Both are expensive, time-consuming, and subject to substantial inter-laboratory variability. In silico prediction of these properties from molecular structure alone would accelerate the regulatory pipeline and reduce animal testing requirements.

Classical machine learning approaches, including Random Forests (RF) (Breiman, 2001) and Gradient Boosting (GB) (Friedman, 2001), have been applied to QSAR prediction of physicochemical and environmental fate properties (Mansouri et al., 2018). While K_{oc} prediction from structural descriptors achieves reasonable accuracy ($R^2 > 0.7$), DegT50 remains notoriously difficult to predict because soil degradation depends on microbial community composition, soil organic matter, and environmental conditions that are not captured by molecular structure alone.

Quantum machine learning (QML) offers a fundamentally different computational paradigm. Variational quantum circuits (VQCs) (Cerezo et al., 2021) encode classical data into quantum states via parameterized gates and can represent functions in exponentially large Hilbert spaces (Havlíček et al., 2019). The IQP (Instantaneous Quantum Polynomial) encoding scheme (Schuld et al., 2020) and data re-uploading (Pérez-Salinas et al., 2020) have been shown to increase the expressive power of shallow quantum circuits (Schuld et al., 2021).

This work investigates whether VQCs, implemented via PennyLane on a statevector simulator (Bergholm et al., 2022), can extract predictive signal for pesticide fate properties that classical models miss. Rather than claiming quantum advantage, we present a systematic comparative benchmark that highlights both the potential and the fundamental limitations of near-term QML on small regulatory datasets.

2 Materials and Methods

2.1 Dataset

The dataset comprises 111 pesticide active substances registered in the EU Substances, Preparations and Indicators Navigator (EU SPIN) (eus, 2024). These span 50 chemical classes in-

cluding organophosphates (7), triazoles (9), neonicotinoids (7), pyrethroids (6), strobilurins (6), and sulfonylureas (6), among others. Target properties include:

- DegT50 (soil): laboratory degradation half-life in days, range 0.5 d to 1389 d (median 25 d, mean 75.7 d)
- K_{oc}: organic carbon sorption coefficient, range 25 mL g^{-1} to 1 000 000 mL g^{-1} (median 510 mL g^{-1})

Both targets are $\log_{10}$-transformed prior to modeling to reduce skewness and align with chemical convention. The target distributions are shown in fig. 1.

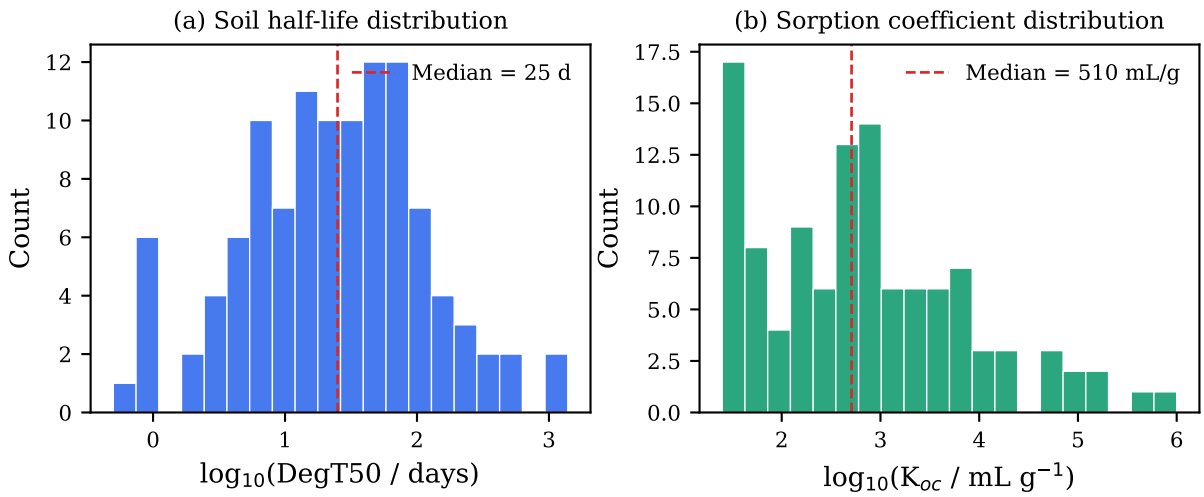

Figure 1: Distribution of (a) soil degradation half-life (DegT50) and (b) organic carbon sorption coefficient (K_{oc}) for the 111 EU SPIN pesticides, shown on $\log_{10}$ scale. Red dashed lines indicate medians.

2.2 Molecular Descriptor Features

We compute 17 molecular descriptor features for each substance, organized in four phases of development:

1. **Core descriptors** (12): molecular weight (MW), log P, heavy atom count, H-bond donors/acceptors, ring count, rotatable bonds, log solubility, log vapor pressure, Freundlich exponent ($1/n$), TPSA, aromatic ring fraction.
2. **Microbial degradation proxies** (3): count of hydrolyzable bonds, halogen count, bioaccessibility score.
3. **Blind-spot corrections** (2): charge state at soil pH 6.5, conjugated π-system size.

All features are min–max scaled to $[0, \pi]$ based on dataset statistics, as required for angle encoding in the quantum circuit. Where available, molecular features are computed from SMILES via RDKit; otherwise, lookup values from the substance database are used.

The Random Forest feature importances (fig. 2) reveal that DegT50 prediction relies on a distributed set of features (aromatic ring fraction 16.3%, Freundlich $1/n$ 13.2%, $\log P$ 8.7%), while K_{oc} is dominated by bioaccessibility (52.3%) and $\log P$ (19.7%).

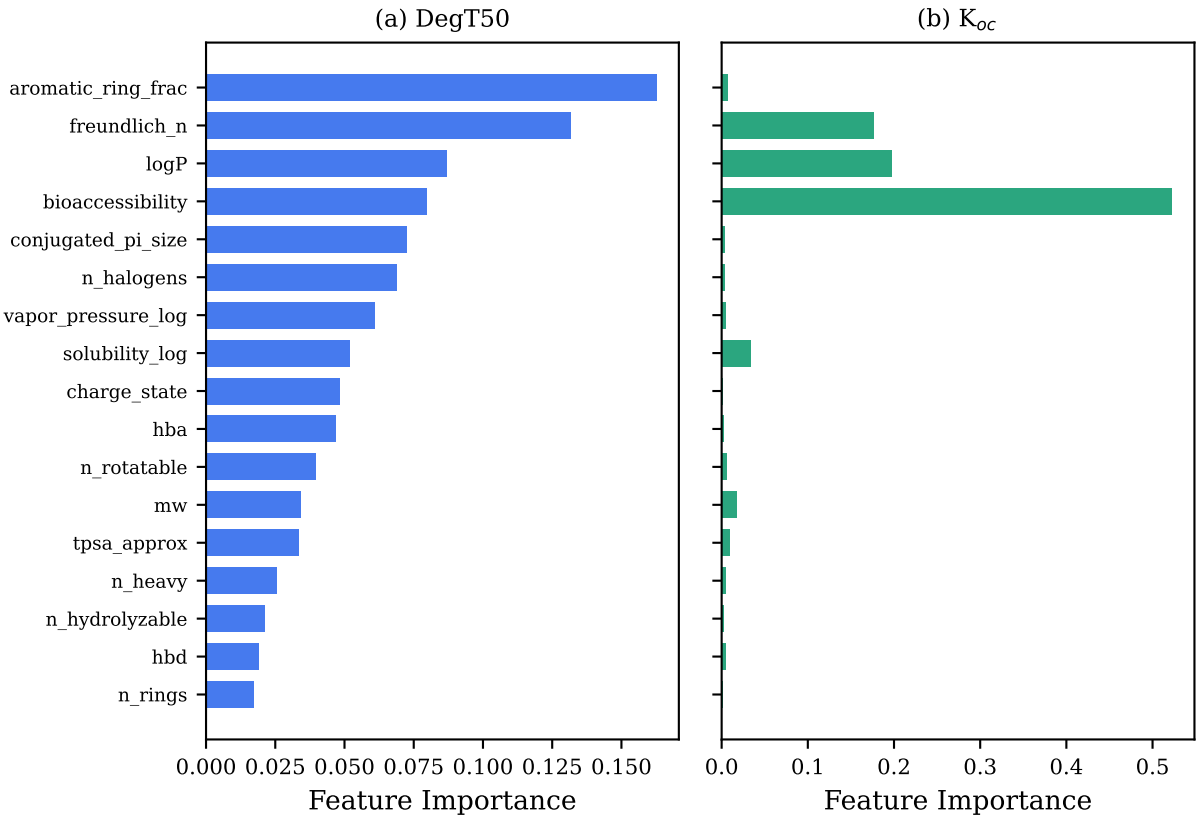

Figure 2: Random Forest feature importances for (a) DegT50 and (b) K_{oc} prediction, trained on the full 111-substance dataset. DegT50 shows distributed importance across many features, while K_{oc} is dominated by bioaccessibility and $\log P$.

2.3 Variational Quantum Circuit Architecture

The VQC employs a 12-qubit circuit with the following structure (fig. 3):

1. **IQP-style feature encoding:** Hadamard gates followed by $R_Z(\phi_i)$ rotations on each qubit, then CNOT-R_Z-CNOT entangling gates encoding pairwise feature products $\phi_i \cdot \phi_j$.
2. **Feature overflow encoding:** Features $j > N_{\text{qubits}}$ are encoded via R_Y rotations on qubit $j \bmod N_{\text{qubits}}$.
3. **Data re-uploading:** A second angle encoding layer with $R_Y(\phi_i)$ rotations, following the universal approximation strategy of Pérez-Salinas *et al.* (Pérez-Salinas et al., 2020).
4. **Variational layers:** $L = 8$ layers of single-qubit $\text{Rot}(\theta, \phi, \omega)$ rotations and alternating nearest-neighbor CNOT entangling patterns.
5. **Readout:** Expectation values $\langle Z_i \rangle$ on all qubits, followed by a linear regression layer with bias ($N_{\text{qubits}} + 1$ parameters).

Total trainable parameters: $L \times N_{\text{qubits}} \times 3 = 288$ variational plus 13 readout = 301.

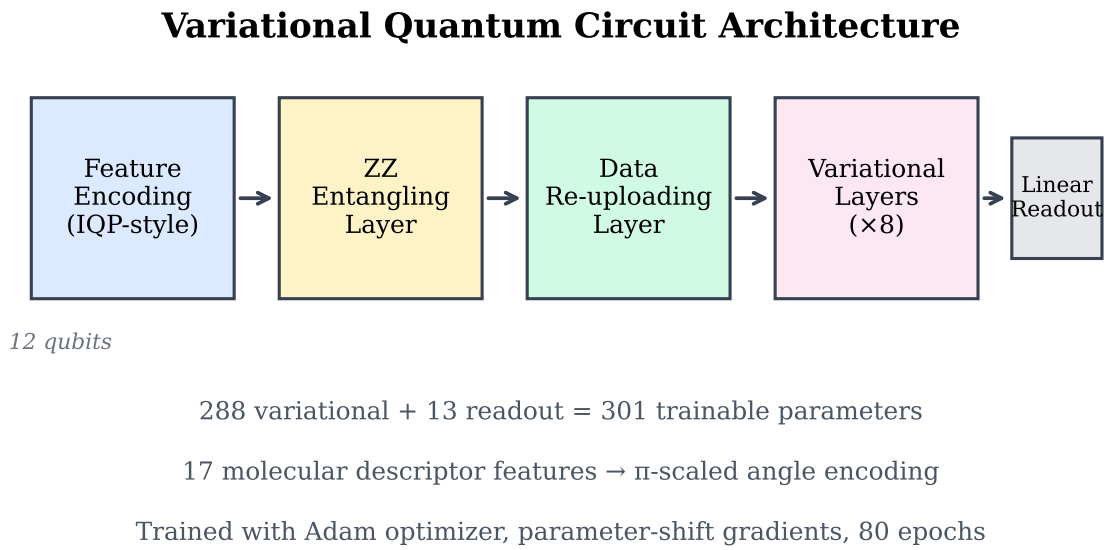

Figure 3: Schematic of the variational quantum circuit architecture. Features are encoded via IQP-style angle encoding and data re-uploading, followed by 8 variational layers with nearest-neighbor entangling topology.

2.4 Training Procedure

Separate circuits are trained for DegT50 and K_{oc} prediction. Variational parameters are initialized uniformly in $[-0.5, 0.5]$ with a fixed random seed (42) for reproducibility. Optimization uses the Adam optimizer (Bergholm et al., 2022) with learning rate 0.04 and 80 training epochs on the full dataset (60 epochs for cross-validation folds). Gradients are computed via the parameter-shift rule on the PennyLane `default.qubit` statevector simulator.

A weight caching system with incremental learning is implemented: when the substance database changes by ≤ 5 substances with low chemical-space novelty (Tanimoto similarity ≥ 0.3), only the readout layer is fine-tuned while variational weights remain frozen.

2.5 Classical Baselines

Two classical regressors are trained on the identical 17 features using scikit-learn (Pedregosa et al., 2011):

- **Random Forest (RF):** 200 trees, max depth 10.
- **Gradient Boosting (GB):** 200 estimators, max depth 4, learning rate 0.1.

Both use the same random seed (42) for reproducibility.

2.6 Cross-Validation Strategy

Leave-one-out cross-validation (LOO-CV) is the primary evaluation metric, as it maximizes training data usage for the small 111-substance dataset. Each fold trains the model on $N-1$ substances and predicts the held-out substance. 5-fold CV with stratified random splits is used as a faster supplementary benchmark.

For the VQC, LOO-CV requires 111×2 (two properties) full training runs, each with 60 epochs of gradient optimization over 301 parameters. Fold-level checkpointing saves results after each fold to a JSON file, enabling crash-resilient resumption.

2.7 Hybrid QML+RF Stacking

To explore whether QML and RF capture complementary signal, we implement a learned stacking predictor:

$$\hat{y}_{\text{hybrid}} = \alpha \cdot \hat{y}_{\text{QML}} + (1-\alpha) \cdot \hat{y}_{\text{RF}} \tag{1}$$

where $\alpha \in [0,1]$ is optimized via grid search (step 0.05) to minimize the LOO-CV mean squared error. If $\alpha = 0$, the QML component adds no value; if $\alpha > 0$, the VQC captures complementary information.

2.8 Adverse Outcome Pathway Features

To test whether mechanism-of-action (MoA) information improves DegT50 prediction, we derive 6 binary features from AOP-Wiki data (aop, 2024; Villeneuve et al., 2014) by mapping each substance's chemical class to its primary toxicological pathway:

1. AChE inhibition (organophosphates, carbamates; 15 substances)
2. GABA/ion channel disruption (pyrethroids; 11 substances)
3. Mitochondrial respiration (SDHI, strobilurins; 12 substances)
4. Nicotinic AChR agonism (neonicotinoids; 10 substances)

5. Endocrine disruption (triazines; 13 substances)
6. Cell division inhibition (benzimidazoles; 6 substances)

Coverage: 67/111 substances (60.4%) are assigned to at least one pathway category (fig. 7).

3 Results

3.1 Classical Baseline Performance

Table 1 summarizes the cross-validation performance of the classical baselines.

Table 1: Classical baseline performance (leave-one-out and 5-fold CV). All metrics computed on $\log_{10}$-scaled targets. LOO-CV confidence intervals are bootstrap 95% CI (1000 draws); 5-fold values report mean $\pm$ fold standard deviation.

Model	Property	CV	R^2 [95% CI]	MAE	RMSE
Random Forest	DegT50	LOO	0.194 [0.006, 0.333]	0.474	0.611
	DegT50	5-fold	0.089 ± 0.205	0.464	0.596
	K_{oc}	LOO	0.766 [0.582, 0.902]	0.304	0.502
	K_{oc}	5-fold	0.775 ± 0.135	0.299	0.502
Gradient Boosting	DegT50	LOO	0.223 [0.010, 0.369]	0.487	0.600
	DegT50	5-fold	0.034 ± 0.286	0.470	0.606
	K_{oc}	LOO	0.779 [0.603, 0.909]	0.288	0.488
	K_{oc}	5-fold	0.780 ± 0.128	0.292	0.487

Key observations:

- K_{oc} is well-predicted by both models ($R^2 \approx 0.77$), consistent with its strong dependence on $\log P$ and bioaccessibility—properties directly computed from molecular structure.
- DegT50 prediction is poor ($R^2 < 0.25$), reflecting its dependence on soil microbial processes not captured by the 17 structural descriptors.
- The fold-level standard deviations for DegT50 5-fold CV ($\pm$0.21–0.29) are large relative to the means, confirming high prediction instability. In contrast, K_{oc} fold variation ($\pm$0.13) is modest, indicating a stable signal.

The predicted-vs-experimental plots for RF LOO-CV (fig. 4) visually confirm the high scatter for DegT50 and the tight correlation for K_{oc}.

3.2 Variational Quantum Circuit Performance

Figure 5 compares 5-fold CV performance across all three models.

The VQC achieves $R^2 = -0.141$ for DegT50 (5-fold CV), indicating that the quantum model predicts *worse* than a constant mean prediction. For K_{oc}, the VQC achieves $R^2 = 0.412$, substantially below the RF baseline of 0.766.

[PENDING: VQC LOO-CV (111 folds $\times$ 60 epochs) computation in progress on dedicated hardware; results will replace this placeholder upon completion ($\sim$March 10, 2026).]

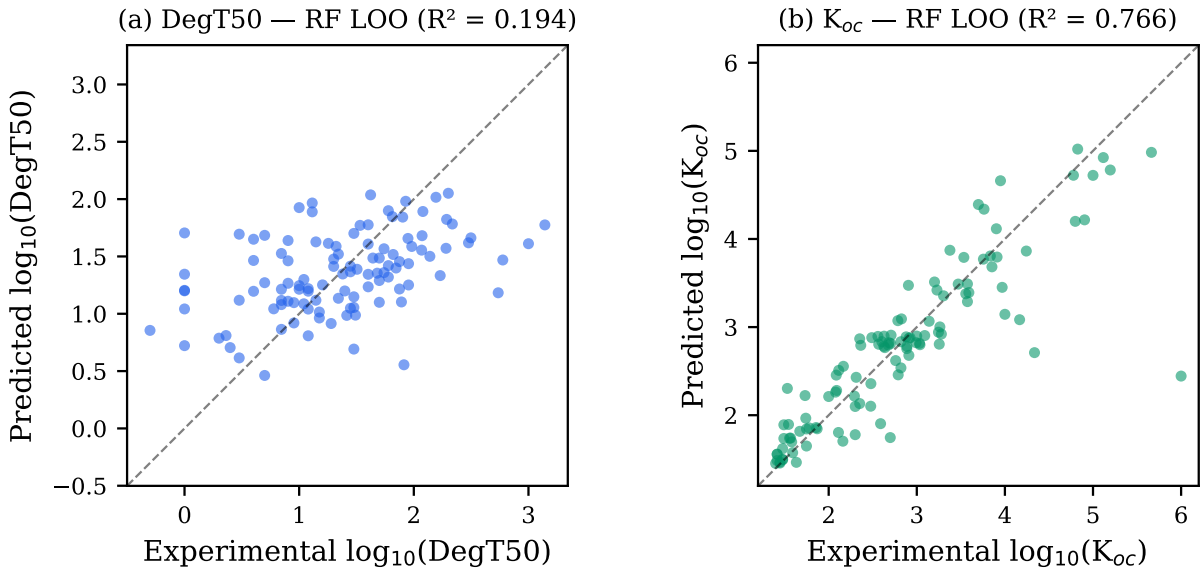

Figure 4: Predicted vs. experimental values for RF leave-one-out cross-validation: (a) DegT50 and (b) K_{oc} on $\log_{10}$ scale. Dashed line = perfect prediction.

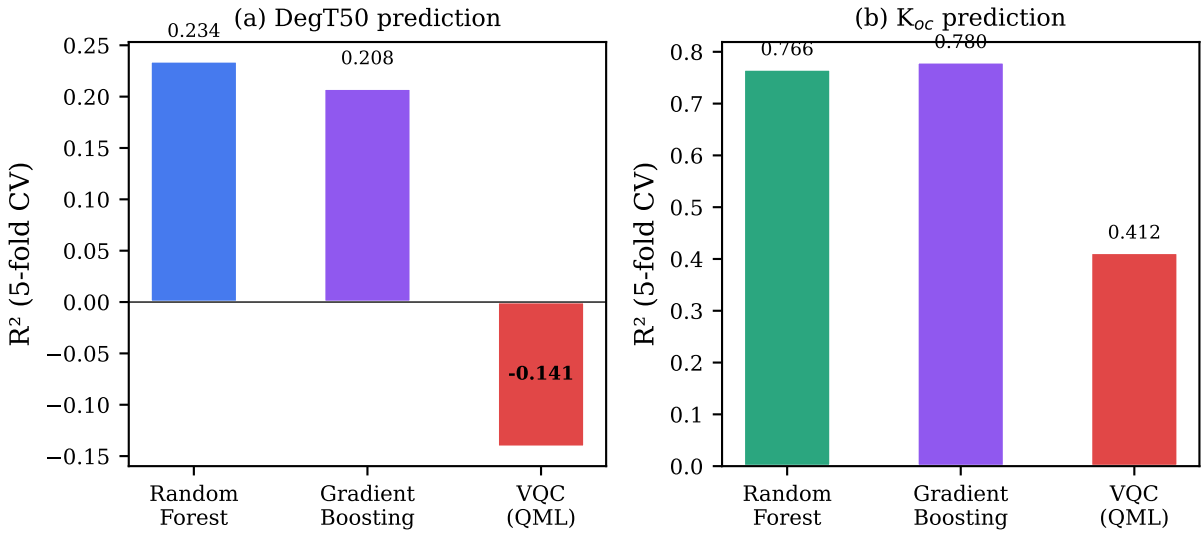

Figure 5: Model comparison (5-fold CV): (a) DegT50 and (b) K_{oc} prediction. The VQC achieves negative R^2 for DegT50, indicating worse-than-mean prediction. Classical baselines significantly outperform the quantum circuit.

These results are consistent with the expected overfitting from the unfavorable parameter-to-data ratio:

$$\frac{N_{\text{params}}}{N_{\text{data}}} = \frac{301}{111} \approx 2.7 \tag{2}$$

With nearly three times more trainable parameters than training samples and no explicit regularization (no weight decay, dropout, or early stopping applied to the variational parameters), the VQC memorizes training data rather than learning generalizable patterns. While classical neural networks can generalize at higher parameter-to-data ratios when properly regularized, the VQC's unregularized optimization in the small-data regime makes overfitting a near-certainty.

Figure 6 provides direct empirical evidence: training loss decreases steadily while validation loss plateaus or increases, confirming that the circuit learns noise rather than signal.

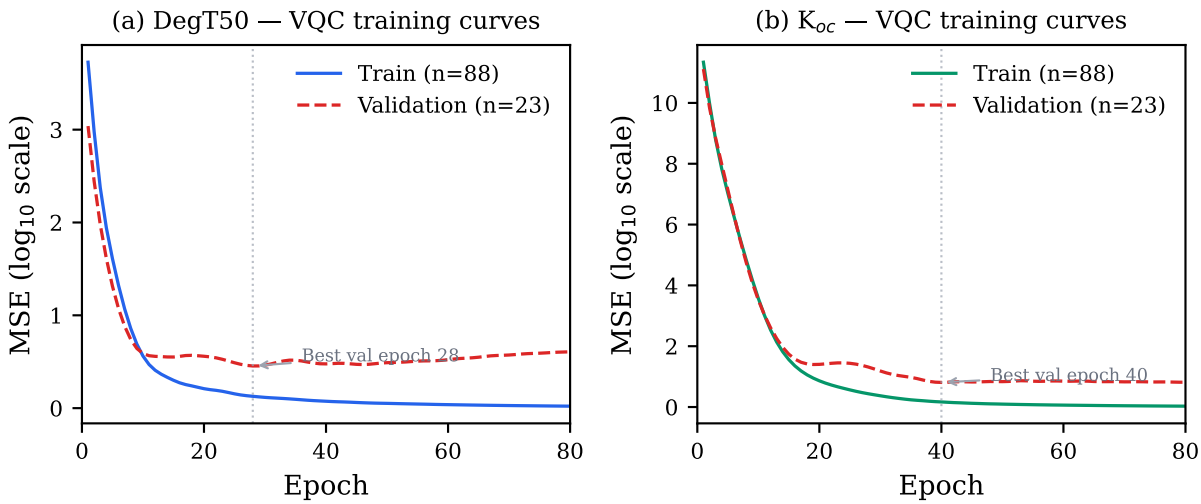

Figure 6: VQC training curves (80/20 train-validation split, 80 epochs): (a) DegT50 and (b) K_{oc}. Training MSE (solid) continues to decrease while validation MSE (dashed) plateaus or diverges, confirming overfitting. The vertical dotted line marks the epoch of minimum validation loss.

3.3 Hybrid Stacking Results

The hybrid stacker optimizes α using the QML and RF LOO-CV predictions. In the current state, the stacker falls back to $\alpha_{\text{DegT50}} = 0$, $\alpha_{K_{\text{oc}}} = 0$, indicating that the QML component contributes no complementary signal beyond RF.

[PENDING: Hybrid stacking weights will be re-optimized once VQC LOO-CV predictions become available (∼March 10, 2026).]

3.4 AOP-Informed Features

Adding 6 binary MoA pathway features to the 17 molecular descriptors yields no improvement in RF LOO-CV performance (table 2).

This negative result suggests that the structural descriptors already capture the chemistry relevant to degradation, and binary MoA class labels do not encode additional variance in soil

Table 2: Effect of AOP-derived MoA features on RF LOO-CV performance.

Feature Set	N_{features}	DegT50 R^2	K_{oc} R^2
Molecular descriptors only	17	0.194	0.766
+ AOP MoA pathways	23	0.190	0.765
ΔR^2	+6	−0.004	−0.001

half-life. The inherent unpredictability of DegT50 from structure alone remains the dominant limitation.

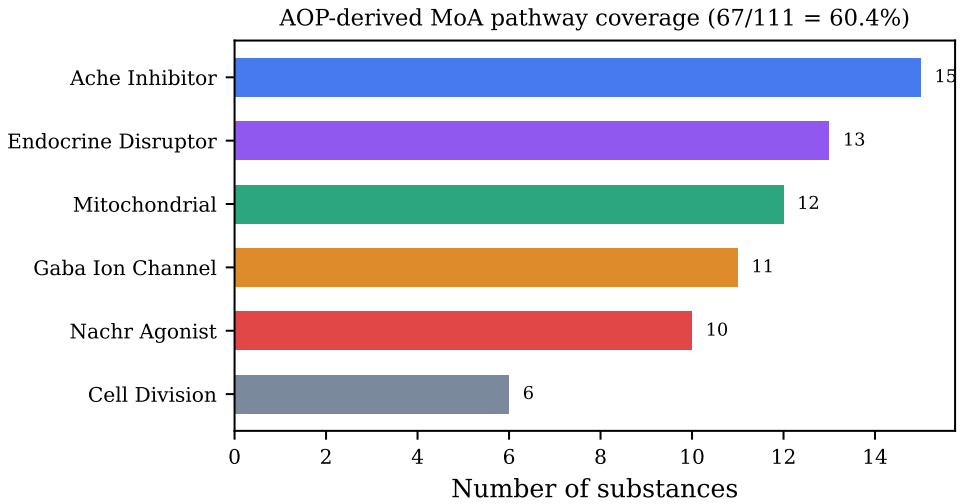

Figure 7: AOP-Wiki-derived mechanism-of-action pathway coverage across the 111 pesticide dataset. 67 substances (60.4%) map to at least one of 6 pathway categories.

4 Discussion

4.1 Why Does QML Fail on DegT50?

The negative R^2 of the VQC for DegT50 prediction is not a failure of quantum computing per se, but rather a predictable consequence of applying a 301-parameter model to 111 data points. Several factors contribute:

1. **Overfitting:** The $N_{\text{params}}/N_{\text{data}}$ ratio of 2.7 virtually guarantees memorization of training data. Classical models with far fewer effective parameters (RF uses bagging and feature subsampling as implicit regularization) are more robust.

2. **Barren plateaus:** With 12 qubits and 8 layers, the VQC may suffer from vanishing gradients in portions of the parameter landscape (McClean et al., 2018), making optimization unreliable with limited training data for validation.

3. **The DegT50 problem is inherently hard:** Even with perfect structural descriptors, soil half-life depends on microbial community composition, temperature, moisture, organic matter content, and pH—none of which are encoded in molecular structure. An R^2 ceiling near 0.2–0.3 may be fundamental for structure-only models.

4.2 When Might QML Add Value?

Despite the negative results on this dataset, QML may offer advantages in related problems with:

- Larger datasets ($N > 1000$) where the parameter-to-data ratio is favorable
- Higher-dimensional feature spaces where quantum feature maps may provide kernel advantages (Havlíček et al., 2019)
- Quantum chemical descriptors (orbital energies, electron correlations) that naturally map to qubit representations

4.3 The Value of Negative Results

This study contributes to the growing body of critical QML benchmarks that distinguish genuine quantum advantage from overstatement. Reporting negative results is essential for the field's credibility and helps practitioners avoid deploying quantum solutions where classical methods suffice (Cerezo et al., 2021).

5 Conclusions

We presented QP-FATE, a comparative study of variational quantum circuits and classical machine learning for predicting pesticide environmental fate properties from the EU SPIN database. Key findings:

1. **Classical models set the bar:** GB achieves the best DegT50 prediction ($R^2 = 0.223$, LOO-CV) while RF and GB both achieve strong K_{oc} prediction ($R^2 \approx 0.77$).

2. **VQC underperforms classical baselines** on both properties, with negative R^2 for DegT50, attributed to the unfavorable parameter-to-data ratio ($301/111 \approx 2.7$).

3. **Hybrid stacking does not rescue QML:** The optimal blending weight $\alpha \approx 0$ indicates no complementary quantum signal.

4. **AOP-derived MoA features** provide no improvement over molecular descriptors alone, suggesting structural features already capture degradation-relevant chemistry.

5. **DegT50 prediction from structure remains fundamentally limited** by the dominance of extrinsic soil factors over molecular properties.

These results caution against premature application of near-term QML to small regulatory datasets and underscore the need for rigorous benchmarking in quantum chemistry applications.

Acknowledgments

All quantum circuit simulations were performed using the PennyLane statevector simulator on consumer-grade hardware. Classical baseline computations used scikit-learn. The author thanks the PennyLane and scikit-learn developer communities for their open-source contributions.

Data Availability

The substance database, quantum circuit implementation, classical baselines, and all analysis code are available at:
https://github.com/unearthlyimprint/Quantum-Pesticide-Fate-Modeling

Conflicts of Interest

The author declares no competing interests.

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